CS340
Project Step 6

Website URL: http://flip3.engr.oregonstate.edu:26801

Team Name: Team 132
Project Title: Latte Larry's Coffee Shop

Team Members:
- Aaron Underhill
- Fernando Valle III

EXECUTIVE SUMMARY:

- In step 1, we were pretty comfortable with the database in mind, so we didn't make any significant changes to the tables/entities; however, we made plenty of small changes based on peer feedback. This included improving the wording for various attributes to improve clarity, specifying the correct data types and limits for different attributes, and adding a few more relevant attributes to the Order-Drinks table.
- After receiving feedback in step 3, we edited our DDL.sql file to allow for a nullable relationship between the Employees and Orders table. This allowed for a more logical and stable relationship between the tables (i.e. an order won't be erased simply because an employee is removed from the database).
- Further changes were made to the SQL file to allow for DELETE ON CASCADE operations.
- We also added CSS to our web page design to improve appearance and readability.
- After step 4, we focused on correctly implementing CRUD steps into our web page. This included incorporating/fixing UPDATE functionalities for the Customers table, as well as fixing the DELETE functionality.At this point, functionality was sufficient, and we were progressing toward incorporating CRUD functionalities into our other tables.
- In step 5, we emphasized correctly implementing CRUD functionality into all tables. However, both team members had personal circumstances that interfered with their goals for this step, so not all CRUD functionalities were properly implemented as originally intended. Still, significant progress was made and CRUD functionality was added to most tables.
- After step 5 and into step 6, we focused strongly on the overall appearance of the design. This included adding CSS to make the overall web page design clearer and visually appealing to the user, improving the wording of texts on the web page, and adding some images to create the sense of a friendly webpage. Unfortunately, both members once again had personal issues that interfered with completing the project, so not all CRUD operations were successfully incorporated into the webpage. Still, the team members put much effort into creating a near-complete product.

PROJECT OUTLINE:

Latte Larry's is a newly created but already well-known coffee shop in a bustling city, with over 400 customers ordering there daily. Due to the demand, the shop currently employs over 10 baristas, and the shop currently averages over $1500 in daily sales. This database-driven website will create an easy-to-see overview of the coffee shop's employees, customers, orders, and drinks. It'll make it easy for the shop owners to view how employees are performing, how often their customers order at the shop, what kind of orders customers are typically getting, and what sort of drinks are most popular throughout the day and make the most in overall sales. On top of that, each employee will have a password associated with them that will be used to access the till in order to improve security. To keep track of all this information in our database, we've made the following tables: Employees, Passwords, Customers, Orders, Drinks, Orders-Drinks.

DATABASE OUTLINE:

- **Employees:** records the details of the Employees that work for out business.
    - idEmployee: INT, auto_increment, unique, not NULL, PK
    - name: VARCHAR(50), no NULL
    - phoneNum: VARCHAR(20), not NULL
    - email: VARCHAR(145), not NULL
    - address: VARCHAR(145), not NULL
    - idPassword: INT, not NULL, FK
    - Relationships:
        - 1:M relationship with Orders – 1 Employee can have served many orders
        - 1:1 relationship with Passwords – 1 Employee will have 1 Password to sign into their till

- **Passwords**: records the Passwords that's associated with each employee that works for our business.
    - idPassword: INT, auto_increment, unique, not NULL, PK
    - password: VARCHAR(256), not NULL
    - Relationships:
        - 1:1 relationship with Employees

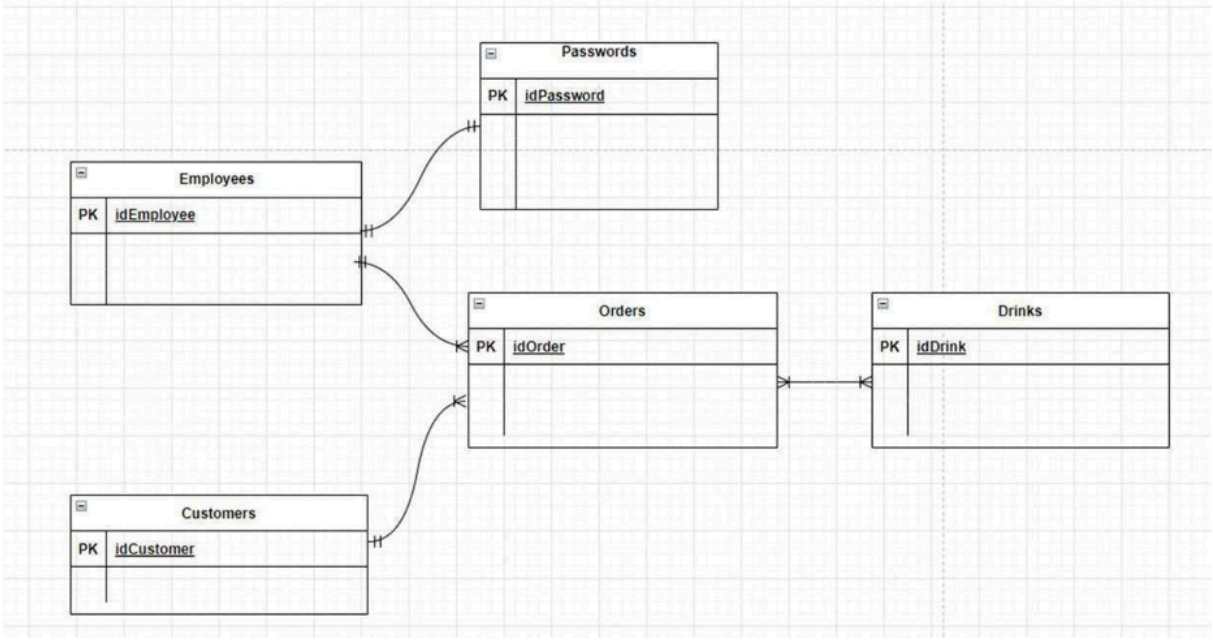
- **Customers**: records the details of the Customers that shop at our coffee shop.
    - idCustomer: INT, auto_increment, unique, not NULL, PK
    - name: VARCHAR(50), not NULL
    - phoneNum: VARCHAR(20), not NULL
    - email: VARCHAR(145), not NULL
    - Relationships:
        - 1:M relationship with Orders – 1 Customer can have many orders

- **Orders**: records the details of customers' Orders placed at our coffee shop.
    - idOrder: INT, auto_increment, unique, not NULL, PK
    - idEmployee: INT, FK, not NULL
    - idCustomer: INT, FK, not NULL
    - totalPrice: DECIMAL(5, 2), not NULL
    - minutesToComplete: DECIMAL(3, 1), not NULL
    - orderDate: DATE, not NULL
    - Relationships:
        - M:1 relationship with Employees
        - M:1 relationship with Customers
        - M:M relationship with Drinks – Many Orders can have many Drinks

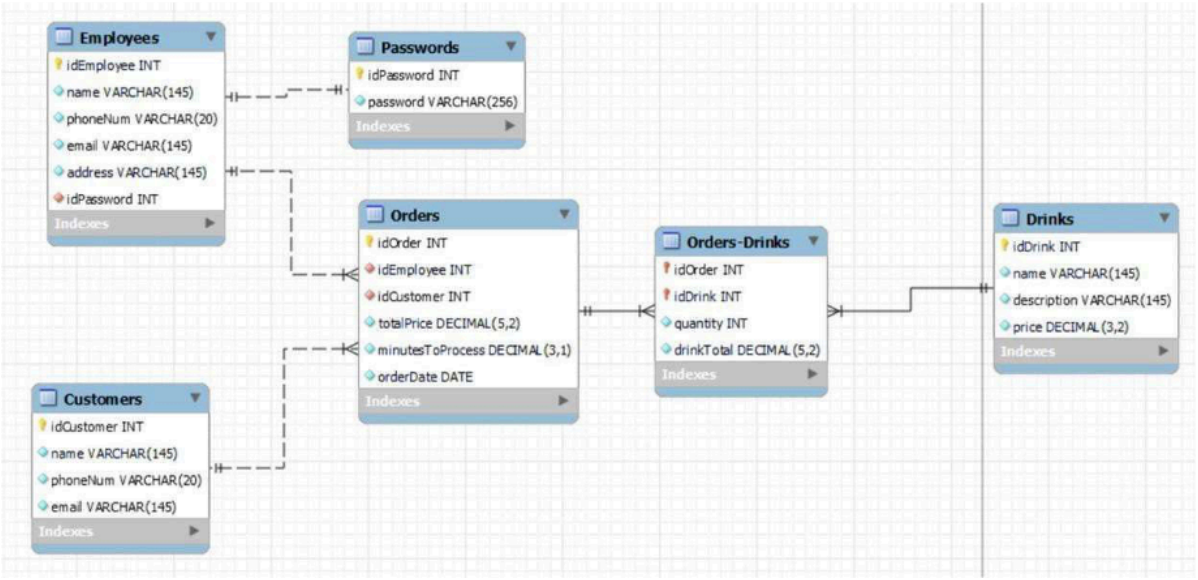
- **Drinks**: records the details of the various Drinks available at our coffee shop.
    - idDrink: INT, auto_increment, unique, not NULL, PK
    - name: VARCHAR(45), not NULL
    - description: VARCHAR(145), not NULL
    - price: DECIMAL(3,2), not NULL
    - Relationships:
        - M:M relationship with Orders

- **Orders-Drinks**: intersection table for Orders and Drinks. Records the quantity of drinks and the total price for the specific drinks.
    - idOrder: INT, not NULL, FK
    - idDrink: INT, not NULL, FK
    - quantity: INT, not NULL
    - drinkTotal: DECIMAL(5,2), not NULL

ERD DIAGRAM:



SCHEMA DIAGRAM:



SAMPLE DATA:

Employees

idEmployee	name	phoneNum	email	address	idPassword
1	John Smith	555-555-5555	John@gmailcom	440 East Knoll	1
2	Sarah Lock	544-444-4444	Sarah@gmail.com	320 Hill Dr	2
3	Adam Hurt	444-333-2222	Adam@hotmail.com	820 Scenic Route	3
4	Susan Sue	333-222-1111	Susan@gmail.com	562 Hill Dr	4
5	Larry David	234-523-2132	Larry@gmail.com	295 David Blvd	5

Customers

idCustomer	name	phoneNum	email
1	Maddy Sane	111-111-1111	maddy@gmail.com
2	Hannah Shirley	233-233-2323	hannah@gmail.com
3	Donald Trump	444-454-4545	donnald@gmail.com

4	David Laver	555-454-5555	maddy@gmail.com
5	Bob Smith	541-982-723	bob@gmail.com

Orders

idOrder	idEmployee	idCustomer	totalPrice	minutesToProcess	orderDate
1	1	2	$10	5	4/18/2024
2	4	1	$11	4	4/18/2024
3	2	3	$9	3	4/19/2024
4	3	4	$9	4	4/19/2024
5	5	2	$5.50	2	4/20/2024

Drinks

idDrink	name	description	price
1	Larry's Latte	Just a typical Latte, made by Larry.	$5.00
2	Larry's Cappuccino	Just a typical Cappuccino, made by Larry.	$5.50
3	Larry's Americano	Just your typical Americano, but made by larry.	$4.00

4	Larry's Cold Brew	Normal Cold Brew, but Larry made it.	$5.00
5	Larry's Mocha	Just a typical Mocha, made by Larry	$4.50

Order_Drinks

idOrder	idDrink	quantity	drinkTotal
1	1	1	$5.00
1	4	1	$5.00
2	2	2	$11
3	3	1	$4.00
3	4	1	$5.00
4	1	1	$5.00
4	3	1	$4.00
5	2	1	$5.50

User Interface Screenshots:

Home Page.

No CRUD steps. It's simply the main page of the website and possesses links to the other pages.



Employees (Create, Read, Update, Delete)

Allows the user to add a new employee to the database using the field below "Add a New Employee." Moreover, the page allows the user to see all the current employees currently working at the business, including their personal data as well. The page also allows the user to update an existing Employee's info by using the field below "Updating an Employee's Info." Lastly, the page allows the user to delete an employee from the database by clicking on the "delete" button that's in the corresponding row of the employee.



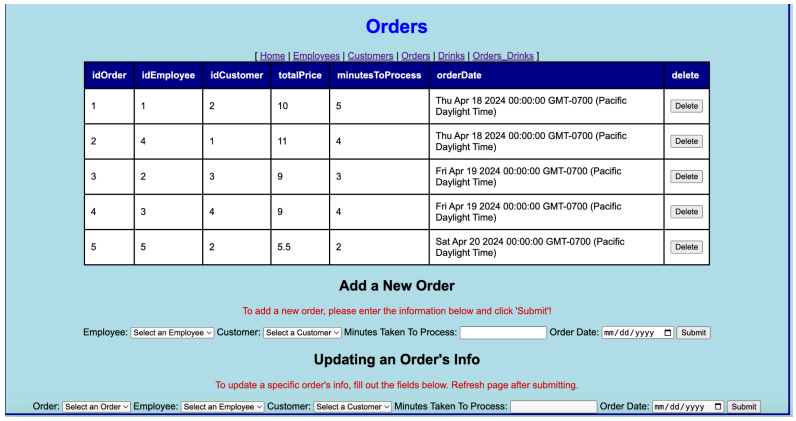
Customers (Create, Read, Update, Delete)

Allows the user to add a new customer to the database using the field below “Add a New Customer.” Moreover, the page allows the user to see all the current customers that have registered as customers at the shop, including their personal data as well. The page also allows the user to update an existing customer’s info by using the field below “Updating a customer’s Info.” Lastly, the page allows the user to delete a customer from the database by clicking on the “delete” button that’s in the corresponding row of the customer. Here, we’ve added DYNAMIC DROPDOWN functionality when updating an existing customer’s information.



Orders (Create, Read, Update, Delete)

Allows the user to add a new order to the database using the field below “Add a New Order.” Moreover, the page allows the user to see all the orders that have been placed at the store, including information such as the order date or the time it took to finish the order. The page also allows the user to update an existing order’s info by using the field below “Updating an order Info.” This was done just in case an order might’ve received the wrong information when created. Lastly, the page allows the user to delete an order from the database by clicking on the “delete” button that’s in the corresponding row of the order. Here, we have a NULLABLE relationship between order and employee. Additionally, we have added DYNAMIC DROPDOWN functionality here, for both adding a new order and updating an existing order.



Drinks (Create, Read, Update, Delete)

Allows the user to add a new drink to the database using the field below “Add a New Drink.” Moreover, the page allows the user to see all the different drinks that are currently offered at the store. The page also allows the user to update an existing drink’s info by using the field below “Updating a Drink’s Info.” This allows the store to change prices, a description, etc. Lastly, the page allows the user to delete a drink from the database by clicking on the “delete” button that’s in the corresponding row of the drink.



Order_Drinks (Create, Read, Update, Delete)

Allows the user to add a new Order_Drink to the database using the field below “Add a New Order_Drink”. Moreover, the page allows the user to see all the different order_drinks that are currently offered at the store. Lastly, the page allows the user to delete an order_drink from the database by clicking on the “delete” button that’s in the corresponding row. Here, we have implemented the delete of a M:N relationship (between orders and drinks). Moreover, we also have the UPDATE of an M:N relationship.

